

# **Aggregate Supply in the Heterogeneous-Agent Model**

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Pascal Michailat  
<https://pascalmichailat.org/w/>

## Aggregate supply

Household  $i$  sells  $y_i = f(x) \times k_i$

Aggregation: all households sell

$$\begin{aligned} y &= \sum_i y_i \\ &= \sum_i f(x) \cdot k_i \\ &= f(x) \sum_i k_i \end{aligned}$$

$$y = f(x) k = y^S(x)$$

$y^S(x)$  aggregate supply curve

↳ AS curve is the same as in the representative - agent model.

Alternative derivation:  $y = m(\sum k_i, \sum v_i)$

$$y = \left( \sum k_i \right) m(1, n)$$

$\uparrow$   $\uparrow$   
 $k$   $f(x)$

$$y = f(x) \cdot k$$